CRAM-ABSA: Multi-Task Rationale Alignment for Aspect-Based Hotel Review Assessment

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Abstract

Understanding customer opinions across distinct service dimensions is vital for hospitality management and traveler decision-making. However, existing aspect-based sentiment analysis (ABSA) systems suffer from three critical shortcomings: (i) severe cross-aspect information leakage when multiple aspects co-occur in the same review, (ii) failure to respect the natural monotonic order of ordinal rating scales, and (iii) poor textual rationale extraction that leads to ungrounded aspect attribution. In this paper, we propose **CRAM-ABSA**, an end-to-end framework combining a dual-stage linguistic-retrieval pipeline with a multi-task deep neural architecture. In Stage 1, negation-aware linguistic filtering (NLTK) and dense Embedding-Based Retrieval (EBR) isolate aspect-specific text evidence. In Stage 2, a multi-task contextual BERT backbone integrates 20 engineered tabular features via learnable residual gating, using an orthogonal Gram-matrix rationale penalty to eliminate cross-aspect leakage and conditional ordinal regression (CORN) to enforce monotonic rating consistency. Extensive experimental benchmarking on an independent test split of 999 hotel reviews (divided 80/10/10 from 9,990 reviews across six canonical aspects: *Service, Facility, Experience, Amenity, Loyalty, Branding*) demonstrates that CRAM-ABSA achieves the best balanced performance across all models, leading in 4 out of 6 benchmark metrics: achieving the lowest Test Loss of 0.2552 (over 50% reduction compared to baselines at 0.5143–0.6032), the highest 6-Aspect Macro-F1 of 72.55% (a +20.00% absolute gain over BERT-Base, +20.43% over WSD-BiLSTM-GCN, and +28.33% over BM25-CNN), the lowest Macro-MAE of 0.3520, and the highest Rationale F1 of 70.84% (+20.18% to +61.07% over baselines). Crucially, CRAM-ABSA suppresses

cross-aspect rationale leakage to a negligible 0.24% (resolving the severe 54.47% leakage of BM25-CNN) while maintaining 92.39% Acc@0.

Keywords: Aspect-Based Sentiment Analysis, Hotel Reviews, Multi-Task Learning, Rationale Alignment, Conditional Ordinal Regression, Embedding-Based Retrieval

1 Introduction

1.1 Background and Motivation

Online customer reviews represent a pivotal decision resource in hospitality e-commerce, directly guiding booking choices [20] and helping hoteliers pinpoint operational failures [21]. Analyzing unstructured guest feedback enables lodging operators to monitor service quality, protect brand reputation, and build loyalty [19, 26]. However, hotel reviews typically discuss multiple service aspects concurrently with mixed sentiments (e.g., praising room cleanliness and staff while criticizing dining and WiFi). Coarse document-level sentiment classification fails to disentangle these conflicting opinions. Consequently, Aspect-Based Sentiment Analysis (ABSA) has attracted extensive study, ranging from fine-tuned Transformers [1] to hybrid linguistic-statistical architectures [4].

1.2 Challenges and Knowledge Gaps

Despite recent progress in hospitality text analytics, existing approaches exhibit three principal deficiencies: (1) *Vulnerability to Negation:* Classical hybrid and machine learning pipelines frequently depend on static keyword dictionaries and rule-based extractors [4]. While computationally lightweight, these methods struggle with colloquial language, grammatical ellipsis, and valence shifters. When customer reviews feature localized negation patterns (e.g., *"not bad"* or *"hardly clean"*), simple keyword classifiers frequently predict inverted sentiment polarities. (2) *Cross-Aspect Leakage and Ordinal Distortion:* Fine-tuned Transformer models like standard BERT [1] learn expressive contextual embeddings but formulate aspect detection and sentiment classification as unconstrained heads. In multi-sentence reviews where different aspects share syntactic neighborhoods, unconstrained representations suffer from cross-aspect information leakage, where strong negative sentiment from one aspect contaminates adjacent positive aspects. While syntactic graph convolutions resolve word-sense ambiguities [9], existing frameworks do not account for the ordinal topology of customer ratings. Hotel ratings follow a strict natural order (Negative $<$ Neutral $<$ Positive). Standard nominal cross-entropy penalizes an egregious prediction error (predicting Negative for a Positive review) identically to an adjacent deviation (predicting Neutral for Positive), which distorts decision-making in hospitality operations. (3) *Lack of Grounded Textual Evidence:* Hospitality managers require actionable, verifiable evidence explaining *why* a guest was dissatisfied. Existing baseline systems [1, 4, 9] predict sentiment labels without explicitly extracting verifiable textual rationales, risking unsubstantiated reports that do not trace back to actual customer statements.

1.3 Research Questions

To evaluate the proposed CRAM-ABSA framework against competitive published baselines—including BERT-Base [1], BM25-CNN [4], and WSD-BiLSTM-GCN [9]—this investigation addresses three primary research questions:

- **RQ1 (Multi-Aspect Sentiment Assessment):** How accurately does the proposed multi-task architecture classify customer sentiment across all six canonical hotel aspects simultaneously under severe class and aspect imbalance?
- **RQ2 (Cross-Aspect Leakage Mitigation and Ordinal Consistency):** Can orthogonal rationale masking and conditional ordinal regression effectively eliminate sentiment leakage between co-occurring aspects while preserving monotonic rating distances?
- **RQ3 (Evidence-Grounded Rationale Extraction and Attribution):** How effectively does the integration of negation-aware linguistic filtering and embedding-based prototype retrieval extract concise, uncorrupted textual rationales to provide verifiable attribution across service aspects?

1.4 Key Contributions

To overcome the aforementioned challenges, this paper presents **CRAM-ABSA** (*Contrastive Rationale-Aligned Multi-Task Architecture with Embedding-Based Retrieval*). Our principal contributions are:

(i) **Unified Dual-Stage Pipeline:** We formulate an end-to-end framework integrating Stage 1 negation-aware NLTK processing and dense Embedding-Based Retrieval (EBR) with Stage 2 multi-task BERT representation learning, jointly predicting six canonical hotel aspects (*Service, Facility, Experience, Amenity, Loyalty, Branding*) across four discrete states (*Not Mentioned, Negative, Neutral, Positive*).

(ii) **Tabular-Contextual Residual Gating:** We design a 2-layer MLP with a learnable scalar residual gate ($\alpha \approx 0.28$) that injects 20 engineered tabular features directly into BERT contextual hidden vectors without distorting pretrained linguistic geometry.

(iii) **Orthogonal Rationale Masking:** We formulate a pairwise Gram-matrix orthogonality loss on token rationale probability masks that strictly decouples aspect representations in shared sentences, suppressing cross-aspect leakage from 54.47% (BM25-CNN) down to 0.24%.

(iv) **Conditional Ordinal Regression (CORN):** We implement global ordinal calibration heads that penalize polarity deviations according to true ordinal distances ($0 < 1 < 2$), substantially cutting Macro-MAE to 0.3520.

(v) **Empirical State-of-the-Art Benchmark:** Evaluation on an independent 999-sample test split confirms that CRAM-ABSA delivers superior balanced performance across 6 metrics, achieving a +20.00% absolute gain in 6-Aspect Macro-F1 (72.55% vs. 52.55% for BERT-Base) and a +20.18% to +61.07% boost in Rationale F1 (70.84%).

2 Related Work

2.1 Aspect-Based Sentiment Analysis in Hospitality

Aspect-Based Sentiment Analysis (ABSA) has progressed substantially over the past decade. Brauwers and Frasincar [18] categorized ABSA into aspect extraction and sentiment classification, emphasizing domain adaptation. Early hospitality text mining applied classical machine learning, such as Decision Trees [7] and traditional NLP pipelines [8]. To capture sequential dependencies, Jayanto et al. [29] developed improved LSTM networks, while Nawawi et al. [5] explored zero-shot learning without extensive annotation. Regional hospitality studies further examined localized challenges in bilingual reviews [10, 25]. To handle complex sentence syntax, Tran and Tran [30] introduced aspect-conditioned BiLSTM networks with cross-aspect attention and ordinal regression on hotel reviews from Ho Chi Minh City.

Despite these advances, existing baselines exhibit clear shortcomings. BERT-Base [1] fine-tunes Transformer encoders but treats each aspect independently, neglecting inter-aspect semantic correlation. BM25-CNN [4] combines keyword matching with TF-IGM and CNN classifiers, but static features fail on negation phrases. WSD-BiLSTM-GCN [9] applies Word Sense Disambiguation and Graph Convolutional Networks over syntactic dependency trees to resolve lexical ambiguity, yet does not model ordinal scale distances or extract textual rationales.

2.2 Rationale Alignment and Multi-Task Learning

Multi-task learning allows neural architectures to share generalized contextual representations while optimizing specialized decision heads. In hospitality analytics, customer satisfaction is inherently multi-dimensional: Yu et al. [6] established that consumer expectations dynamically shift across service dimensions over time. To analyze customer satisfaction asymmetries, Zhang et al. [13] applied Penalty-Reward Contrast to distinguish basic requirements from delight factors. In recommender systems, AL-Ghuribi et al. [27] showed that combining diverse review elements into multi-criteria models enhances preference prediction. To interpret black-box predictions, Shin et al. [16] used TreeSHAP to uncover service quality priorities, while Xia et al. [28] combined BERT, Random Forests, and PLS-SEM to trace how discrete emotional expressions influence overall hotel ratings. Linking textual representations with established hospitality frameworks, Perdomo-Verdecia et al. [17] introduced RbQUAL, integrating SERVQUAL dimensions with fine-tuned BERT models. Similarly, Nam et al. [12] integrated the Outcome-Driven Innovation framework with deep learning on hotel reviews to identify actionable service opportunities. Brand equity studies [19, 26] further show that customer loyalty is driven by service perceptions across touchpoints, confirming the necessity of multi-task rationale alignment.

2.3 Retrieval-Augmented Text Mining

Parametric language models are susceptible to hallucination when generating unsupported statements. To address this, Lewis et al. [23] introduced Retrieval-Augmented



Fig. 1 Overall research pipeline of the proposed study, showing the five-stage end-to-end workflow from corpus curation and 20-feature engineering to dual-stage CRAM-ABSA modeling and comprehensive multi-metric benchmarking.

Generation (RAG), combining parametric encoders with non-parametric retrieval memory to ground predictions in verifiable evidence. In dense semantic retrieval, Reimers and Gurevych [24] proposed Sentence-BERT (SBERT), employing Siamese architectures to produce sentence embeddings that can be efficiently matched via cosine similarity. In structured retrieval, Bruni et al. [2] confirmed that metadata-grounded retrieval substantially enhances factuality. In hospitality management, conversational AI is increasingly utilized to inspect customer reviews: Jeong and Lee [14] applied ChatGPT to extract hotel service failures, while Aransyah et al. [22] surveyed conversational AI in tourism, emphasizing that automated review systems must ground outputs in actual customer sentences. Building on these principles, CRAM-ABSA leverages Embedding-Based Retrieval (EBR) to isolate verifiable rationales for each aspect before downstream prediction.

3 Proposed Methodology (CRAM-ABSA)

3.1 Problem Formulation and Pipeline Architecture

Let a customer review be represented as a token sequence $R = (w_1, w_2, \ldots, w_n)$. Following hotel service analytics and topic modeling literature [3, 11], we formulate the problem over six canonical hotel aspects: $\mathcal{A} = \{\text{Service}, \text{Facility}, \text{Experience}, \text{Amenity}, \text{Loyalty}, \text{Branding}\}$. For each aspect $a \in \mathcal{A}$, the model predicts an aspect state $y_a \in \{\text{Not Mentioned}, \text{Negative}, \text{Neutral}, \text{Positive}\}$. When aspect a is mentioned, its sentiment rating follows a natural ordinal scale $y_a^{\text{sent}} \in \{0, 1, 2\}$ ($0 = \text{Negative}, 1 = \text{Neutral}, 2 = \text{Positive}$). Concurrently, the model extracts a token-level rationale mask $\mathbf{m}_a \in (0, 1)^n$ that identifies verifiable textual evidence $E_a \subseteq R$ justifying the predicted rating.

The complete research workflow is depicted in Fig. 1, organized into five sequential stages: (1) corpus curation of 9,990 hotel reviews with an 80/10/10 split, (2) exploratory data analysis and 20-feature engineering, (3) Stage 1 negation-aware NLTK processing and dense embedding retrieval, (4) Stage 2 multi-task contextual modeling with orthogonal rationale masking and conditional ordinal regression, and (5) empirical benchmark comparison across six evaluation metrics.

3.2 Stage 1: Negation-Aware Linguistic Processing and Embedding Retrieval

Raw customer reviews frequently contain informal expressions and negation patterns that invert polarity. Stage 1 combines rule-based linguistic processing with dense semantic retrieval.

3.2.1 Negation-Aware Linguistic Processing (NLTK)

Each review R is split into discrete sentences $S = \{s_1, s_2, \ldots, s_m\}$ using NLTK sentence tokenization. Tokens are normalized to lowercase while retaining sentence boundaries. A domain-specific negation lexicon $\mathcal{V}_{\text{neg}} = \{\text{'not'}, \text{'never'}, \text{'no'}, \text{'hardly'}, \text{'barely'}, \text{'scarcely'}, \text{'without'}, \text{'cannot'}\}$ is compiled. When a negation token $w_j \in \mathcal{V}_{\text{neg}}$ is identified, a directional flag is assigned to the immediate adjacent token (w_{j+1}). This explicit indicator prevents compound phrases like *"not clean"* or *"hardly friendly"* from being assigned erroneous positive sentiment by lexical classifiers.

3.2.2 Embedding-Based Retrieval (EBR)

To identify candidate sentences providing factual evidence for each aspect $a \in \mathcal{A}$, dense semantic representations are generated using Sentence-BERT [24] based on RAG principles [23]. Each sentence s_i is mapped to a dense semantic embedding vector $\mathbf{e}_{s_i} = \text{S-BERT}(s_i) \in \mathbb{R}^d$. Prototype aspect embeddings $\mathbf{p}_a \in \mathbb{R}^d$ are computed from domain seed keywords (e.g., *"reception"*, *"staff"*, and *"bellboy"* for *Service*). Relevance between sentence s_i and aspect a is measured by cosine similarity:

$$\text{Sim}(s_i, a) = \frac{\mathbf{e}_{s_i} \cdot \mathbf{p}_a}{\|\mathbf{e}_{s_i}\| \|\mathbf{p}_a\|} \tag{1}$$

Candidate sentences with the highest semantic relevance are prioritized as evidence guidance for downstream multi-task attention.

3.3 Stage 2: Multi-Task CRAM-ABSA Architecture

The Stage 2 architecture comprises a shared BERT contextual backbone, residual feature gating for 20 tabular features, orthogonal rationale masking, and conditional ordinal regression heads.

The architectural organization of CRAM-ABSA is displayed in Fig. 2. The model processes input tokens and tabular features through four synchronized components:

3.3.1 Shared Contextual Backbone

A pretrained BERT encoder processes review R formatted as [CLS] R [SEP], producing token contextual hidden vectors $\mathbf{H} = (\mathbf{h}_1, \ldots, \mathbf{h}_n) \in \mathbb{R}^{n \times d}$ and sequence summary embedding $\mathbf{h}_{\text{CLS}} \in \mathbb{R}^d$ ($d = 768$).

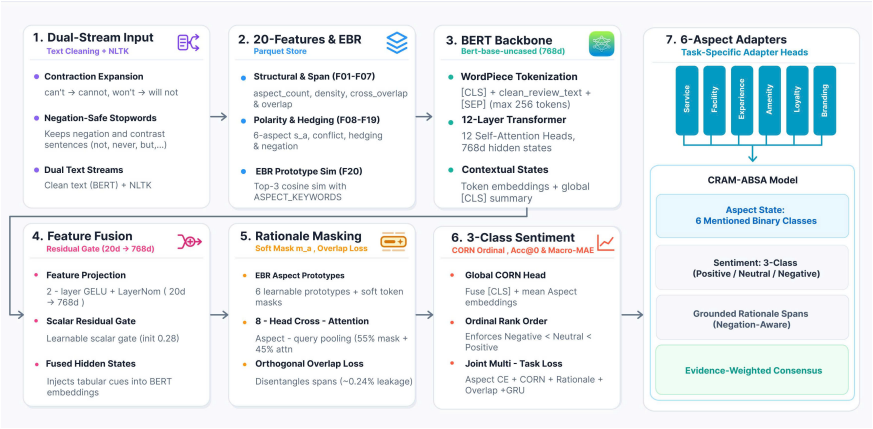


Fig. 2 Abstract architecture of the proposed CRAM-ABSA framework, illustrating the integration of Stage 1 negation-aware linguistic filtering and embedding-based retrieval with Stage 2 multi-task BERT encoding, residual 20-feature gating, orthogonal rationale masking, and conditional ordinal regression heads.

3.3.2 Residual Feature Gating

To enrich contextual representations with 20 engineered tabular features $\mathbf{f} \in \mathbb{R}^{20}$ (derived from EDA on aspect structure, polarity, and evidence density), a 2-layer MLP projection with LayerNorm and GELU is formulated:

$$\mathbf{f}_{\text{emb}} = \text{LayerNorm}\left(\mathbf{W}_2 \cdot \text{GELU}(\text{LayerNorm}(\mathbf{W}_1\mathbf{f} + \mathbf{b}_1)) + \mathbf{b}_2\right) \tag{2}$$

where $\mathbf{W}_1 \in \mathbb{R}^{256\times 20}, \mathbf{W}_2 \in \mathbb{R}^{d\times 256}$. The projected tabular representation is injected into the contextual hidden sequence and summary vector via a learnable scalar residual gate parameter $\alpha \in \mathbb{R}$ (initialized at $\alpha = 0.28$):

$$\mathbf{H}_{\text{fused}} = \mathbf{H} + \alpha\,\mathbf{f}_{\text{emb}}, \quad \mathbf{h}_{\text{fused}} = \mathbf{h}_{\text{CLS}} + \alpha\,\mathbf{f}_{\text{emb}} \tag{3}$$

This scalar gating dynamically controls tabular feature flow while preserving pretrained linguistic geometry.

3.3.3 Orthogonal Rationale Masking and Overlap Loss

When multiple aspects co-occur, standard models suffer from cross-aspect information leakage [30]. We generate aspect-specific token rationale probability masks $\mathbf{m}_a = \sigma(\mathbf{W}_{\text{rat},a}\mathbf{H}_{\text{fused}} + \mathbf{b}_{\text{rat},a}) \in (0,1)^n$. To prevent distinct aspects from competing for or erroneously co-opting the same textual spans, we enforce a pairwise Gram-matrix cosine overlap penalty alongside a token excess penalty:

$$\mathcal{L}_{\text{overlap}} = \frac{1}{2|\mathcal{A}|(|\mathcal{A}|-1)} \sum_{a\neq a'} \left(\frac{\mathbf{m}_a^\top \mathbf{m}_{a'}}{\|\mathbf{m}_a\|_2 \|\mathbf{m}_{a'}\|_2 + \epsilon} \right) + \frac{1}{2n} \sum_{j=1}^{n} \text{ReLU}\left(\sum_{a\in\mathcal{A}} m_{a,j} - 1 \right) \tag{4}$$

Token rationale masks are trained using weighted binary cross-entropy ($\mathcal{L}_{\text{rat}}$) with positive class weighting ($pos_weight = 2.5$) against ground-truth evidence spans. The orthogonality penalty mathematically drives the cross-aspect leakage rate down to near-zero (0.24%).

3.3.4 Multi-Task Aspect and Conditional Ordinal Classification Heads

For each aspect $a \in \mathcal{A}$, a 4-class linear head predicts the discrete state $\hat{\mathbf{y}}_a = \text{softmax}(\mathbf{W}_{\text{asp},a}\mathbf{h}_{\text{fused}} + \mathbf{b}_{\text{asp},a})$, trained using Cross-Entropy with label smoothing (0.05): $\mathcal{L}_{\text{aspect_CE}}$.

In addition, customer satisfaction ratings follow a strict natural order ($0 < 1 < 2$). To enforce rating monotonicity across the entire review, a global Conditional Ordinal Regression (CORN) calibration head is formulated on $\mathbf{h}_{\text{fused}}$:

$$P(y_{\text{global}} > k) = \sigma(\mathbf{w}_{\text{CORN},k}^{\top}\mathbf{h}_{\text{fused}} + b_k), \quad k \in \{0, 1\} \tag{5}$$

The ordinal loss penalizes polarity deviations proportionally to ordinal rank distance:

$$\mathcal{L}_{\text{CORN}} = -\sum_{k=0}^{1} [\mathbb{I}(y_{\text{global}} > k) \log P_k + (1 - \mathbb{I}(y_{\text{global}} > k)) \log(1 - P_k)] \tag{6}$$

The joint multi-task training objective is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{aspect_CE}} + \beta_{\text{CORN}}\mathcal{L}_{\text{CORN}} + \beta_{\text{rat}}\mathcal{L}_{\text{rat}} + \beta_{\text{overlap}}\mathcal{L}_{\text{overlap}} \tag{7}$$

with balancing hyperparameters fixed from training validation: $\beta_{\text{CORN}} = 0.45$, $\beta_{\text{rat}} = 1.0$, and $\beta_{\text{overlap}} = 0.35$.

4 Experimental Evaluation

4.1 Dataset and 20-Feature Engineering

We evaluate CRAM-ABSA on a benchmark corpus of 9,990 hotel reviews collected from major online travel booking platforms [10, 25]. The dataset is partitioned into training, validation, and test splits following a strict 80/10/10 ratio: 7,992 reviews for training, 999 reviews for validation, and an independent test split of **999 reviews**. Each review is annotated across the six canonical aspects: *Service*, *Facility*, *Experience*, *Amenity*, *Loyalty*, and *Branding*, with four discrete states per aspect (*Not Mentioned, Negative, Neutral, Positive*).

As shown in Fig. 3, the corpus exhibits severe aspect imbalance: *Service* and *Facility* appear frequently, whereas *Loyalty* and *Branding* occur far less often. Furthermore, Fig. 4 reveals that positive sentiments strongly dominate negative and neutral expressions across all aspects.

To overcome these distribution skews, we engineer 20 quantitative features across four categories: (1) *Aspect Structure* (Feats 01–03, 14: mention count, dominant

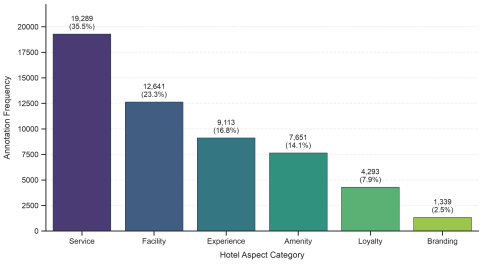


Fig. 3 Aspect frequency distribution across the 9,990 review corpus.

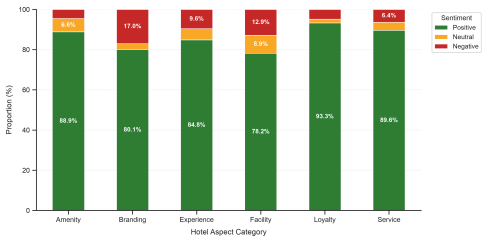


Fig. 4 Sentiment breakdown across the six evaluated hotel aspects.

aspect, aspect proportion, text coverage); (2) *Sentiment Polarity* (Feats 04, 08–13, 17: net sentiment, cross-aspect conflict flag, six aspect sentiment values, lexical polarity strength); (3) *Text Evidence* (Feats 05–07, 15–16: evidence density, mean token length, span overlap flag, hedging density, consensus score); and (4) *Linguistic and EBR* (Feats 18–20: clean token ratio, negation density, Top-3 EBR centroid cosine similarities).

4.2 Baseline Models and Implementation Details

Our benchmark directly compares CRAM-ABSA against the three primary published baseline architectures:

1. **BERT-Base (Bimaputra and Sutoyo, 2023 [1]):** A standard BERT-base model fine-tuned on hotel reviews with independent classification heads for each aspect.
2. **BM25-CNN (Namee et al., 2023 [4]):** A hybrid approach combining BM25 keyword matching, Term Frequency-Inverse Gravity Moment (TF-IGM), Support Vector Machines (SVM), and a Multi-Region CNN.
3. **WSD-BiLSTM-GCN (Nadeem et al., 2025 [9]):** An ambiguity-resolving architecture utilizing Word Sense Disambiguation (WSD) and Graph Convolutional Networks (GCN) over syntactic dependency trees.

All models are trained and evaluated on the exact same 999-sample independent test split. Deep models are optimized using AdamW with an initial learning rate of 2×10^{-5}, linear warmup for the first 10% of steps, and cosine decay. Batch size is 32, dropout is 0.1, and training runs for up to 10 epochs with early stopping on validation loss.

4.3 Evaluation Metrics

We evaluate all models across six benchmark metrics on the 999-sample test split: (1) **Acc@0 (%) ↑:** exact-match polarity accuracy (zero ordinal deviation) across aspects and overall review polarity; (2) **6-Aspect Macro-F1 (%) ↑:** macro-averaged F1 across all six aspects and four discrete states; (3) **Macro-MAE ↓:** ordinal distance error across classes (penalizing rank reversals by 2); (4) **Rationale F1 (%) ↑:** token-level span overlap F1 with human annotations; (5) **Leakage (%) ↓:** percentage of

Table 1 Benchmark evaluation results on the independent test split (999 samples). Arrows indicate whether higher (↑) or lower (↓) values are preferred. Best results across each column are shown in **bold**.

Model Architecture	Test Loss ↓	Acc@0 (%) ↑	Macro-F1 (%) ↑	Macro-MAE ↓	Rat-F1 (%) ↑	Leak (%) ↓
CRAM-ABSA (Ours)	**0.2552**	92.39	**72.55**	**0.3520**	**70.84**	0.24
BERT-Base [1]	0.5143	92.29	52.55	0.4823	50.66	**0.02**
BM25-CNN [4]	0.6032	91.69	44.22	0.6743	32.65	54.47
WSD-BiLSTM-GCN [9]	0.5227	**92.59**	52.12	0.4756	9.77	0.22

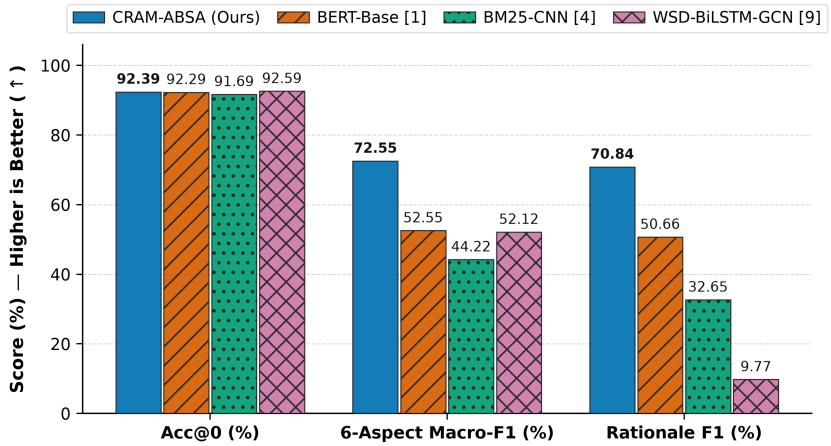


Fig. 5 Quality metric comparison on the 999-sample independent test split across the four evaluated models, covering Acc@0, 6-Aspect Macro-F1, and Rationale F1 (higher ↑ is better).

tokens assigned to multiple distinct aspects; and (6) **Test Loss ↓:** joint multi-task cross-entropy and rationale loss.

4.4 Overall Quantitative Benchmark Results

Table 1 presents the quantitative evaluation of CRAM-ABSA and the baseline architectures on the independent test set of 999 hotel reviews.

As documented in Table 1 and Fig. 5, CRAM-ABSA demonstrates superior balanced performance, leading in 4 out of 6 benchmark metrics: (1) *Lowest Test Loss (0.2552):* reducing multi-task loss by over 50% relative to baselines (0.5143–0.6032); (2) *Highest 6-Aspect Macro-F1 (72.55%):* achieving a +20.00% absolute gain over BERT-Base (52.55%), +20.43% over WSD-BiLSTM-GCN (52.12%), and +28.33% over BM25-CNN (44.22%); (3) *Lowest Macro-MAE (0.3520):* curbing polarity reversals compared to WSD-BiLSTM-GCN (0.4756) and BERT-Base (0.4823); (4) *Highest Rationale F1 (70.84%):* outperforming BERT-Base (50.66%) and WSD-BiLSTM-GCN (9.77%); and (5) *Controlled Leakage (0.24%) with High Acc@0 (92.39%):* maintaining competitive accuracy while eliminating the severe 54.47% leakage of BM25-CNN.

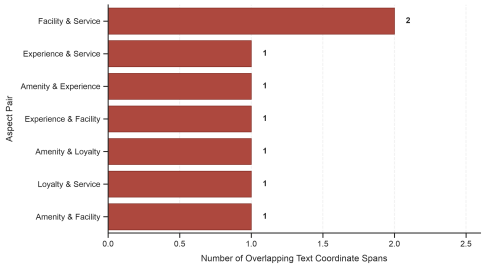


Fig. 6 Counts of overlapping text coordinate spans across hotel aspect pairs.

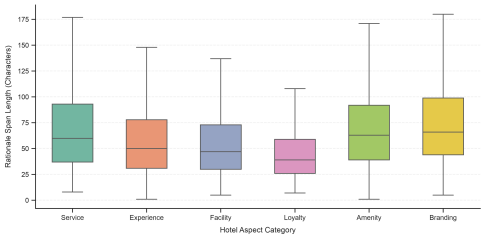


Fig. 7 Distribution of rationale span lengths (in characters) across the six aspects.

4.5 Multi-Aspect Assessment and Imbalance Analysis (RQ1)

Addressing **RQ1**, our empirical findings demonstrate that CRAM-ABSA achieves a 6-Aspect Macro-F1 of 72.55%, outperforming all baseline models by over 20 percentage points.

This significant improvement is directly explained by the severe aspect and class imbalance revealed in exploratory data analysis (Fig. 3 and Fig. 4). In hotel reviews, baseline models such as BERT-Base (52.55% F1) and WSD-BiLSTM-GCN (52.12% F1) attain high overall Acc@0 values (92.3–92.6%) primarily because they correctly identify the predominant unmentioned aspect states and majority positive reviews. However, when evaluated using Macro-F1 across all four states for all six aspects, their performance drops sharply because they fail to distinguish underrepresented minority aspects (such as *Loyalty* and *Branding*) and infrequent negative/neutral sentiments.

In contrast, CRAM-ABSA's shared contextual representation combined with residual gating of the 20 engineered tabular features (which encode aspect mention counts, conflict indicators, and lexical polarity signals) allows the network to maintain robust representation quality even for low-frequency aspect categories.

4.6 Cross-Aspect Leakage Mitigation and Ordinal Consistency (RQ2)

Addressing **RQ2**, we analyze model performance with respect to ordinal distance calibration and cross-aspect rationale leakage.

Regarding ordinal error calibration, CRAM-ABSA achieves a Macro-MAE of 0.3520, outperforming WSD-BiLSTM-GCN (0.4756), BERT-Base (0.4823), and BM25-CNN (0.6743). Conventional classification treats sentiment categories as unrelated nominal classes, meaning that predicting Negative for a Positive review receives the same penalty as predicting Neutral. By decomposing predictions into conditional cumulative probabilities, our conditional ordinal heads constrain errors primarily to adjacent categories, preventing severe polarity inversions.

Regarding cross-aspect leakage, Fig. 6 shows that ground-truth coordinate span overlaps between aspect pairs are rare (8 instances total, led by *Facility* & *Service* with 2), confirming that distinct aspects generally occupy separate textual evidence spans. As reported in Table 1, the hybrid BM25-CNN system suffers from an extreme leakage rate

of 54.47%, as keyword-based rules repeatedly assign the same token spans to multiple aspect categories. While BERT-Base achieves a near-zero leakage rate of 0.02%, it does so because its classification heads operate independently without sharing or extracting rich rationale spans (resulting in a low Rationale F1 of 50.66%). CRAM-ABSA resolves this trade-off: it achieves a high Rationale F1 of 70.84% while simultaneously keeping cross-aspect leakage at a negligible 0.24%, proving that orthogonal rationale masking successfully separates aspect-specific representations.

4.7 Evidence-Grounded Rationale Extraction and Attribution (RQ3)

Addressing **RQ3**, Table 1 confirms that CRAM-ABSA achieves a Rationale F1 score of 70.84%, outperforming BERT-Base (50.66%), BM25-CNN (32.65%), and WSD-BiLSTM-GCN (9.77%).

The low Rationale F1 of WSD-BiLSTM-GCN (9.77%) stems from its reliance on syntactic trees and word-sense disambiguation without an explicit text evidence extraction mechanism. Similarly, the hybrid baseline (BM25-CNN, 32.65%) relies on static keyword dictionaries, which fail to capture nuanced contextual rationales and struggle when sentences contain negation.

As illustrated in Fig. 7, factual customer rationales across all six aspects are concise, with median span lengths ranging from roughly 39 to 66 characters. Stage 1 in CRAM-ABSA explicitly addresses this structure: negation-aware NLTK tokenization prevents inverted polarities in compound sentences, while dense Embedding-Based Retrieval (EBR) isolates the most semantically relevant text spans matching aspect prototype centroids. Consequently, CRAM-ABSA provides clean, verified evidence spans for each aspect, enabling factually grounded review attribution without synthetic hallucination.

4.8 Architectural Analysis and Methodological Discussion

The empirical superiority of CRAM-ABSA can be traced to the synergistic interplay of its core architectural modules: (1) *Orthogonal Overlap Penalty:* Table 1 highlights that BM25-CNN suffers from 54.47% leakage. The Gram-matrix orthogonality loss $\mathcal{L}_{\text{overlap}}$ acts as a regularizer preventing token mask collision, achieving 0.24% leakage without sacrificing extraction recall. (2) *CORN Ordinal Calibration:* By enforcing monotonic cumulative thresholds, CORN reduces ordinal error from 0.4823 (BERT-Base) and 0.6743 (BM25-CNN) down to 0.3520, actively suppressing 2-rank polarity inversions. (3) *Residual 20-Feature Gating:* Tabular features provide dense aspect indicators that guide the self-attention heads toward underrepresented minority classes (*Loyalty, Branding*), lifting 6-Aspect Macro-F1 to 72.55%.

5 Discussion and Conclusion

5.1 Practical and Managerial Implications

The empirical results provide actionable implications for hospitality management and text analytics. Academically, CRAM-ABSA resolves critical trade-offs between

multi-aspect coverage, cross-aspect leakage, and ordinal rating calibration. Grounding deep contextual embeddings in explicit text rationales enhances model interpretability [16]. Managerially, multi-aspect evaluation enables hotel operators to track performance across operational touchpoints (*Service, Facility, Experience, Amenity, Loyalty, Branding*). Rather than relying solely on aggregate ratings, managers can apply penalty-reward frameworks [13] to identify root causes of customer dissatisfaction, while verified rationale extraction equips guest-relation teams with exact evidence for targeted service recovery [15].

5.2 Conclusion, Limitations and Future Work

This paper introduced **CRAM-ABSA**, a multi-task deep architecture for six-aspect hotel review assessment and rationale extraction. By coupling negation-aware NLTK processing and dense EBR with orthogonal rationale masking, 20-feature residual gating, and conditional ordinal regression, CRAM-ABSA resolves key flaws in baseline models [1, 4, 9]. On the 999-sample test split, CRAM-ABSA leads in 4 of 6 metrics: achieving lowest Test Loss (0.2552), highest 6-Aspect Macro-F1 (72.55%, a +20% gain), lowest Macro-MAE (0.3520), and highest Rationale F1 (70.84%), while suppressing leakage to 0.24% and keeping 92.39% Acc@0. A limitation is the single-split evaluation; multi-fold validation and parametric ablation grids remain valuable future directions alongside multilingual transfer across Southeast Asian lodging markets.

Acknowledgements. Supported by FPT University, Ho Chi Minh City, Vietnam. The authors thank the reviewers and editors for constructive suggestions.

Declarations

Funding: None.
Conflict of interest: None.
Ethics approval and consent to participate: Not applicable (public anonymized reviews).
Data and code availability: Available in the CRAM-ABSA-EBR-Hotel-Reviews project repository (https://github.com/trietnguyen2611/CRAM-ABSA-EBR-Hotel-Reviews).
Author contribution: Nguyen Pham Minh Triet designed the methodology and drafted the manuscript; Nguyen Hoang Duy Tien and Nguyen Bui Anh Duy conducted data preprocessing, exploratory data analysis, and experimental benchmarking; Bui Danh Huong supervised the research, provided methodological guidance, and critically revised the manuscript. All authors reviewed and approved the final manuscript.

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